Data classification and visualisation

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- classifies and displays data using a variety of graphical representations **MA4-DAT-C-01**

Related Life Skills outcomes: MALS-DAT-01

Content

Classify data as either numerical (discrete or continuous) or categorical (nominal or ordinal) variables

- Define a variable in the context of statistics as any characteristic, number or quantity that can be measured or counted
- Classify and describe variables as numerical or categorical
- Describe a numerical variable as either discrete or continuous

  Example(s):

  Discrete (number of students, shoe size) and continuous (height, weight).

  IRD4
- Describe a categorical variable as nominal or ordinal

  Example(s):

  Nominal (blue, green, yellow) and ordinal (small, medium, large).

  IRD3
- Distinguish between and compare numerical (discrete or continuous) and categorical (nominal or ordinal) variables
  IRD4

Display data using graphical representations relevant to the purpose of the data

- Represent single datasets using graphs, including frequency histograms and polygons, dot plots, stem-and-leaf plots, divided bar graphs, column graphs, line graphs, sector graphs and pictograms, with or without digital tools

  Example(s):

  Using data from various sources, examine population characteristics of Aboriginal and/or Torres Strait Islander Peoples across all states and territories. Use various data representations to highlight changes over time.

  IRD3, IRD4

- Include sources, titles, labels and scales when displaying data in a graph

Example(s):

The source indicates where the data was collected, the title introduces the data in the graph, the labels identify each axis and scales identify the possible values of a variable along an axis.

IRD3, IRD4

- Select the type of graph best suited to represent various single datasets and justify the choice of graph

Example(s):

Comparing the characteristics of histograms and column graphs and justifying their use for datasets.

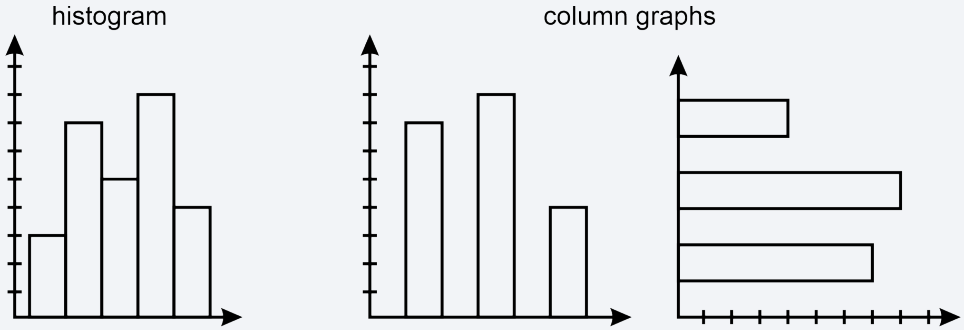


Image long description: A histogram graph with dashes along the vertical axis and the bars close together. 2 column graphs, one with dashes along the vertical axis and another with dashes along the horizontal axis. The bars follow the direction of the dashes and have spaces between them.

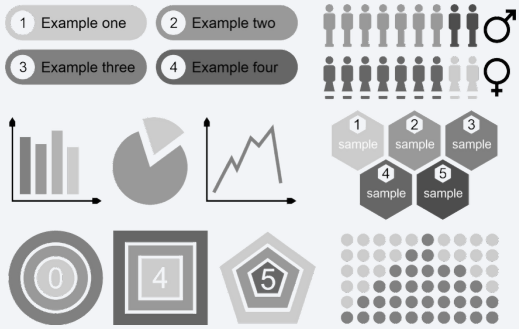
IRD3, IRD4

- Represent a dataset using a statistical infographic and justify the choice of graphical representation used

Example(s):

Creating a statistical infographic showing information about Aboriginal and/or Torres Strait Islander Peoples. Information could include data from the most recent census indicating changes over time in the Aboriginal and Torres Strait Islander population.

Using a statistical infographic to enhance retention and recall of information through visually compelling charts.



IRD6

Interpret data in graphical representations

- Identify and interpret data displayed on graphs

Example(s):

Identifying variables from a histogram, dot plot, stem-and-leaf plot, bar graph, line graph, sector graph and pictogram, interpret the data.

IRD3, IRD4

- Identify features of graphical representations to draw conclusions
  IRD6
- Interpret patterns in graphical representations to make predictions
  IRD6

- Explain why a given graphical representation can lead to a misinterpretation of data

Example(s):

Examining the graphical representation of the cost of services over many years and concluding that the increase in cost of services in Year 4 is much greater than that in Year 3 due to a larger prism.

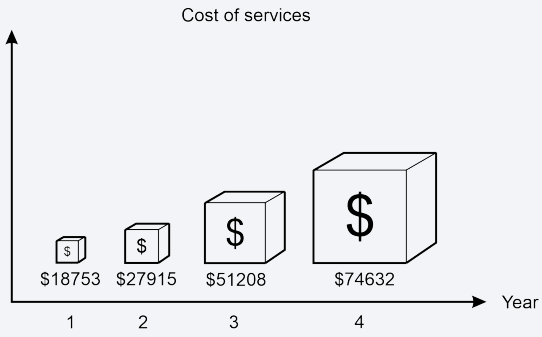


Image long description: Example of a misleading graph depicting cost of services plotted against time in years. Three-dimensional columns have non-uniform values and continue to increase in size as time progresses.

IRD4

Data analysis

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- analyses simple datasets using measures of centre, range and shape of the data **MA4-DAT-C-02**

Related Life Skills outcomes: MALS-DAT-02

Content

Calculate and compare the mean, median, mode and range for simple datasets

- Calculate the mean $(\bar{x})$ of a set of data using digital tools

  Example(s):

  Using the statistical functions of a spreadsheet or another digital tool to determine the mean for both small and large datasets.

  IRD5
- Calculate and describe the mean, median, mode and range of a dataset
  IRD5
- Classify the mean, median and/or mode as measure(s) of centre to represent the average or typical value of a dataset
  IRD5
- Describe and interpret data displays using mean, median and range

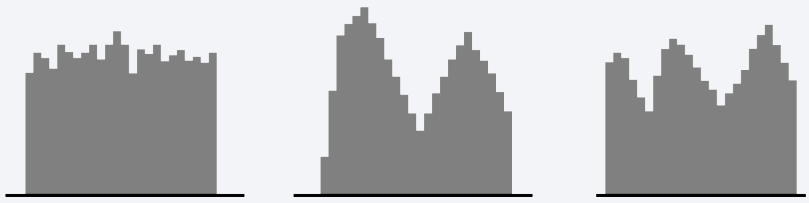
  Example(s):

  Calculating measures of centre and range and using these to describe and interpret data represented in a variety of statistical displays, including frequency distribution tables, histograms, stem-and-leaf plots and dot plots.

  IRD5
- Identify and describe datasets as having no modes (uniform), one mode (unimodal), 2 modes (bimodal) or multiple modes (multimodal)

  Example(s):

  Identifying which of the following diagrams is closest to a uniform distribution:



- Identify the range as a measure of spread to describe variation in a dataset
  IRD5
- Compare simple datasets using the mean, median, mode and range
  IRD5

Interpret the effect individual data points have on measures of centre and range

- Informally identify clusters, gaps and outliers in datasets and give reasons for their occurrence in the context of the data

  Example(s):

  Identifying and describing outliers informally from statistical displays and exploring the reasons why these might occur.

  IRD6
- Identify and explain the impact of adding or removing data values that are clustered at one end of a dataset on the measures of centre

  Example(s):

  Justifying why adding a score of 12 to the set of scores 2, 3, 4, 5, 5, 5, 6, 8 will not change the mode of 5.

  Explaining the effect on the mean if a score of 99% is added to a set of 20 test scores with an average of 60%.

  IRD6
- Identify and explain the impact of outliers on the measures of centre and range

  Example(s):

  Analysing why it is more appropriate to use the median than the mean when the data contains one or more outliers.

  IRD6
- Determine and justify the most appropriate measure of centre to summarise the data in its context

  Example(s):

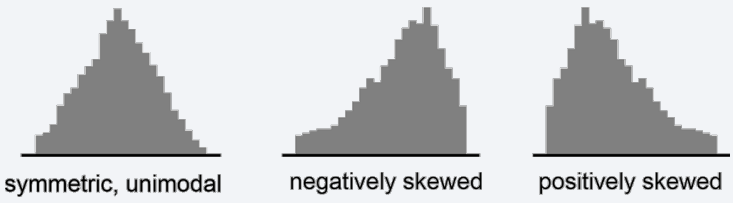
  Using the median as the appropriate measure of centre to determine the average house price in a suburb to avoid distortion of the data by outliers. Using the mean to determine the average test score in a mathematics test when most of the marks are close together.

  IRD6

Analyse datasets presented in various ways and draw conclusions

- Identify and describe the shape and distribution of a dataset using the terms *symmetrical*, *negatively skewed* and *positively skewed*

  Example(s):



- Define a census as a study of every unit, everyone or everything in a population
- Define a sample as a subset of units in a population selected to represent all units in a population of interest

IRD7

- Draw conclusions and make informed decisions about data gathered using data-collection techniques, including census and sampling, which is then presented in tables, graphs and charts

Example(s):

Identifying and describing factors that may have masked results, analysing the accuracy of measurements taken, and determining whether the outcomes of the study can be generalised to other situations.

IRD6, IRD7